

Layered Architecture



Presentation Manager

Presentation Logic

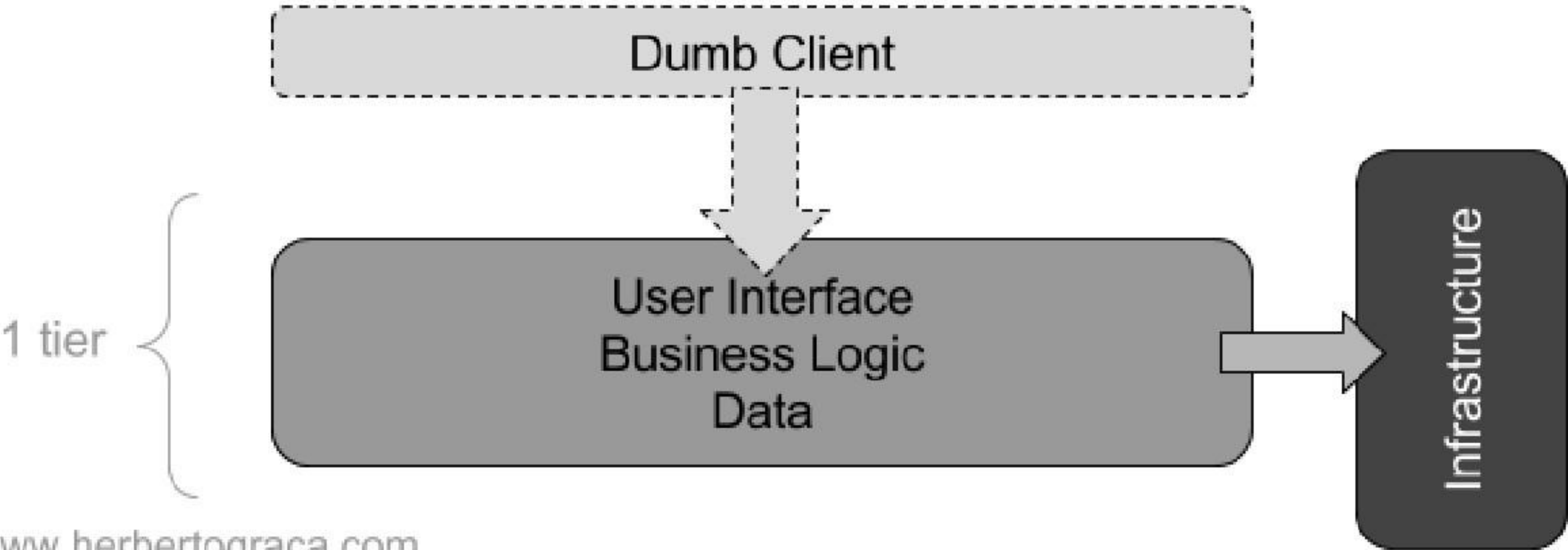
Application Logic

Business Logic

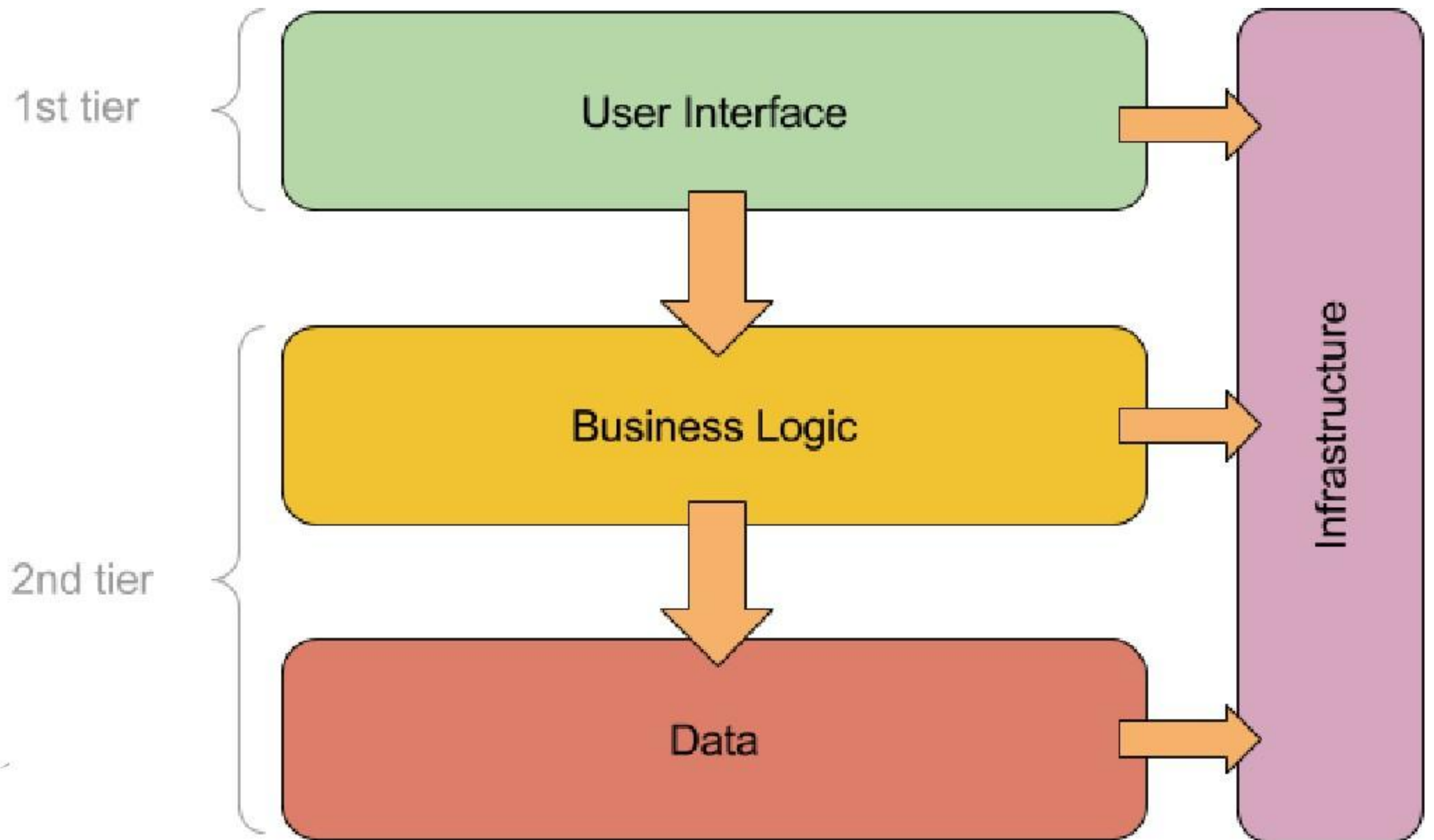
Database Logic

Database Manager

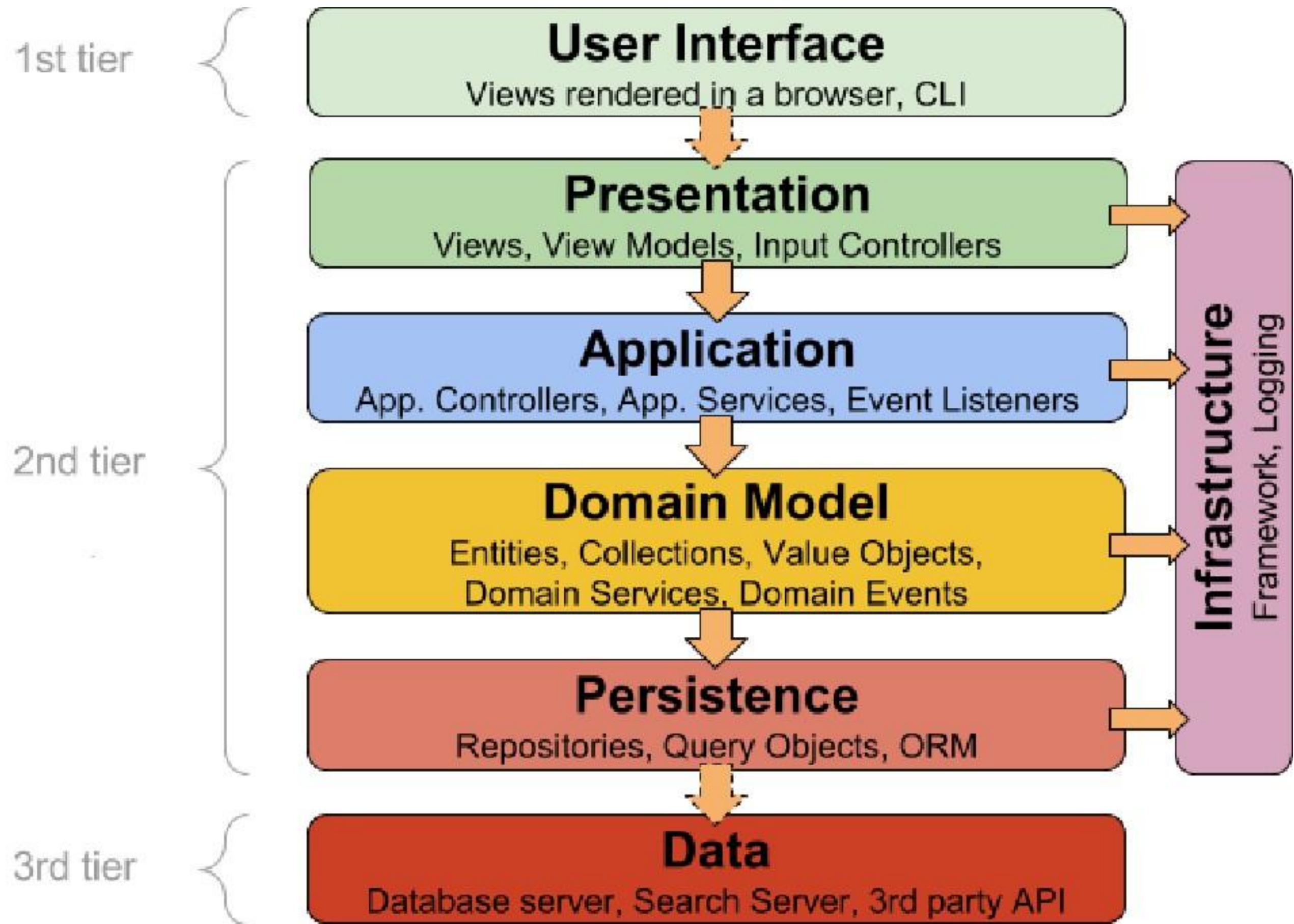
**Single Tier
Application**



2 Tier Application



3 Tier Application



Software Development Layers: Purpose, Responsibilities, Examples, and Technologies

Layer	Purpose	Responsibilities	Examples	Technologies
1. Presentation Manager	Manages the overall user interface environment and user interaction	• Manage windows/screens/pages • Control navigation • Manage UI components • Maintain user sessions • Handle UI resources	• Opening Login Window • Switching from Login Page to Dashboard • Managing menus/toolbars	• JavaFX Stage • Java Swing JFrame • Android Activity • Windows Forms • Browser UI
2. Presentation Logic	Controls the behavior and logic of the user interface	• Handle button clicks • Validate user input • Format data for display • Show messages/errors • Pass requests to application layer	• Check username field is empty • Enable/disable Submit button • Display student marks in table	• JavaFX Controller • Swing ActionListener • JavaScript • React Components • Angular Components • ASP.NET MVC View
3. Application Logic	Controls application workflow and coordinates operations	• Manage application flow • Receive requests from UI • Call business services • Control security and transactions • Coordinate different modules	• Login process • Student registration process • Online shopping checkout process	• Spring Boot Controller/Service • ASP.NET Controller • REST API • Node.js Express • Laravel Controller
4. Business Logic	Implements real-world rules and decisions of the organization	• Apply business rules • Perform calculations • Validate operations • Make decisions • Process data	• Calculate employee salary • Check student attendance eligibility • Calculate bank interest • Apply discount rules	• Java Service Classes • C# Business Classes • Python Classes • Spring Services • .NET Services
5. Database Logic	Converts application requests into database operations	• Create database queries • Insert/update/delete records • Map objects with tables • Retrieve required data • Hide database details from business layer	• Save student record • Search employee by ID • Update customer details	• JDBC • Hibernate • JPA Repository • Entity Framework • DAO • ORM
6. Database Manager	Manages physical database storage and execution	• Store data permanently • Execute SQL commands • Manage connections • Handle transactions • Maintain security and backup • Optimize queries	• Store student table • Execute SELECT query • Maintain database files	• MySQL Server • Oracle Database • SQL Server • PostgreSQL • MongoDB